Program: **Diploma in Integrated Circuit Design and Fabrication**	
Course Code: **4371**	Course Title: **VLSI Design**
Semester: **4**	Credits: **4**
Course Category: **Program core**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the concept of MOS physics, CMOS logic circuit and design
- To understand the MOSFET structure, scaling and fabrication
- To understand the VLSI design methodologies

Course Prerequisites:

Topic	Course code	Course name	Semester
Concept of MOS physics		Solid state devices	3
IC Fabrication Techniques		Introduction to IC fabrication	3
Digital Logic Circuits	3044	Digital electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the basic MOS physics	13	Understanding
CO2	Develop the layout of CMOS logic circuits and explain VLSI design methodologies	15	Understanding
CO3	Demonstrate the realization of CMOS Logic and storage cells	15	Understanding
CO4	Demonstrate the realization of various arithmetic circuits and explain various reliability and testing	15	Understanding

	methods in VLSI		
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic MOS physics		
M1.01	Explain the basic concepts of MOS capacitor	3	Understanding
M1.02	Explain the structure of MOSFET	2	Understanding
M1.03	Describe various short channel effects in MOSFET	4	Understanding
M1.04	Explain MOSFET capacitance	4	Understanding
Contents: Ideal MOS Capacitor, Energy band diagram at equilibrium, accumulation, depletion and inversion, threshold voltage, body effect, MOSFET -Structure, types, Drain current equation (equation only)-linear and saturation region, Drain characteristics, transfer characteristics. Short channel effects (Basic concepts only)-channel length modulation, Drain induced Barrier Lowering, velocity saturation, threshold voltage variations and Hot carrier Effects MOSFET Capacitances-Oxide related capacitance, junction capacitances			
CO2	Develop the layout of CMOS logic circuits and explain VLSI design methodologies		

M2.01	Explain CMOS inverter	3	Understanding
M2.02	Summarize VLSI design flow	4	Understanding
M2.03	Illustrate MOSFET Fabrication	4	Understanding
M2.04	Develop the Layout of CMOS Inverter, NAND and NOR circuits	4	Understanding
	Series Test 1	1	

Contents: CMOS inverter-Switching characteristics, Noise margin, propogation delay, static and dynamic power dissipation

VLSI Design Methodologies-Introduction to Moore's law, ASIC-types of ASIC, SoCs, FPGA devices, VLSI Design flow-Design specifications, Behavioral level RTL Logic Design and Physical level design(basics concept only)

MOSFET Fabrication Techniques and layout-Twin Tub Fabrication sequence, fabrication process flow, Layout Design an Design rules, Stick Diagram and Design rules-micron rules and Lambda rules(Definition only),layout of CMOS inverter, two input NAND and NOR gates

CO3	Demonstrate the realization of CMOS Logic and storage cells		
M3.01	Explain various Static CMOS Logic Design techniques	3	Understanding
M3.02	Explain various Dynamic CMOS Logic Design techniques	4	Understanding
M3.03	Explain the realization of ROM arrays based on NOR and NAND logic	4	Understanding
M3.04	Explain the realization of SRAM and DRAM	4	Understanding

Contents: Static CMOS Logic Design-Realization of logic functions with static CMOS logic, Pass transistor logic and transmission gate logic.

Dynamic Logic Design and storage cells-Pre-charge-Evaluate logic, Domino Logic,NP domino logic, Read Only Memory-4x4 MOS ROM cell Arrays(OR,NOR,NAND),Random Access Memory-SRAM-Six transistor CMOS SRAM cell, DRAM-Three transistor and one transistor Dynamic memory cell

CO4	Demonstrate the realization of various arithmetic circuits and explain various reliability and testing methods in VLSI		
M4.01	Explain the realization of various adder circuits	4	Understanding
M4.02	Explain the realization of Multipliers	4	Understanding

M4.03	Explain Sense amplifiers	3	Understanding
M4.04	Outline various reliability and testing methods in VLSI	4	Understanding
	Series Test 2	1	
Contents: Arithmetic Circuits-Adders-Static Adder, Carry-By pass adder, Linear carry-select adder, square root carry-select adder, multipliers-Array multiplier **Sense Amplifiers-**Basic principles of sense amplifiers, differential sense amplifiers **Reliability and testing of VLSI Circuits-**General concept, MOS testing, test generation methods-Built-In-Self-Test (BIST) and Automatic Test Pattern Generation (ATPG)			

Text / Reference:

T/R	Book Title/Author
T1	Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis & Design, McGraw-Hill, Third Ed., 2003
T2	Wayne Wolf ,Modern VLSI design, Third Edition, Pearson Education,2002
R1	Neil H.E. Weste, Kamran Eshraghian, Principles of CMOS VLSI Design- A Systems Perspective, Second Edition. Pearson Publication, 2005
R2	Razavi - Design of Analog CMOS Integrated Circuits,1e, McGraw Hill Education India Education, New Delhi, 2003
R3	S.M. SZE, VLSI Technology, 2/e, Indian Edition, McGraw-Hill,2003
R4	Suresh Babu V, Solid State Device Technology, Sanguine, 2005.

Online Resources:

Sl.No.	Website Link
1	http://ece-research.unm.edu/jimp/vlsi/slides/chap5_2.html
2	https://www.elprocus.com/cmos-inverter/#:~:text=CMOS
3	https://www.rohm.com/electronics-basics/transistors/understanding-mosfet-characteristics